Fizika

9 sinf

MAVZU: Masalalar yechish

1	Xonaning o‘lchamlari	5x3x2,5 (m)
2	Xonadagi havoning temperaturasi	25℃
3	Xonadagi havoning bosimi	102100 Pa
4	Xonaning hajmi	
5	Xonadagi havoning zichligi	
6	Xonadagi gaz molekulalarining konsentratsiyasi	
7	Xonadagi gaz molekulalarining soni	
8	Xonadagi havoning massasi	

$$\mathrm{V} = \mathrm{a} \cdot \mathrm{b} \cdot \mathrm{c} = 5\,m \cdot 3\,m \cdot 2{,}5\,m = 37{,}5\,m^3$$

$$\mathrm{T} = 25℃ + 273 = 298\,K \qquad M = 0{,}029\,kg/mol$$

$$R = 8{,}31\,\frac{J}{K \cdot mol} \qquad k = 1{,}38 \cdot 10^{-23} J/K$$

$$\rho = \frac{\mathrm{p} \cdot \mathrm{M}}{\mathrm{R} \cdot \mathrm{T}} = \frac{102100\,Pa \cdot 0{,}029\,kg/mol}{8{,}31\,\frac{J}{K \cdot mol} \cdot 298\,K} \approx 1{,}2\,\frac{kg}{m^3}$$

$$n = \frac{p}{k \cdot T} = \frac{102100\ Pa}{1{,}38 \cdot 10^{-23} \frac{J}{K} \cdot 298\ K} \approx 2{,}5 \cdot 10^{25} m^{-3}$$

$$N = n \cdot V = 2{,}5 \cdot 10^{25} m^{-3} \cdot 37{,}5\ m^3 = 9{,}375 \cdot 10^{26}$$

$$m = \rho \cdot V = 1{,}2 \frac{kg}{m^3} \cdot 37{,}5\ m^3 = 45\ kg$$

1	Xonaning o‘lchamlari	5x3x2,5 (m)
2	Xonadagi havoning temperaturasi	25℃
3	Xonadagi havoning bosimi	102100 Pa
4	Xonaning hajmi	$37{,}5\ m^3$
5	Xonadagi havoning zichligi	$1{,}2\ \frac{kg}{m^3}$
6	Xonadagi gaz molekulalarining konsentratsiyasi	$2{,}5 \cdot 10^{25} m^{-3}$
7	Xonadagi gaz molekulalarining soni	$9{,}375 \cdot 10^{26}$ ta
8	Xonadagi havoning massasi	45 kg

Ideal gaz holat tenglamasi

$$\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2}.$$

$$\frac{pV}{T} = const.$$

$$pV = \frac{m}{M} RT$$

$$T = \frac{pVM}{mR}$$

$$p = \frac{mRT}{MV}$$

$$V = \frac{mRT}{Mp}$$

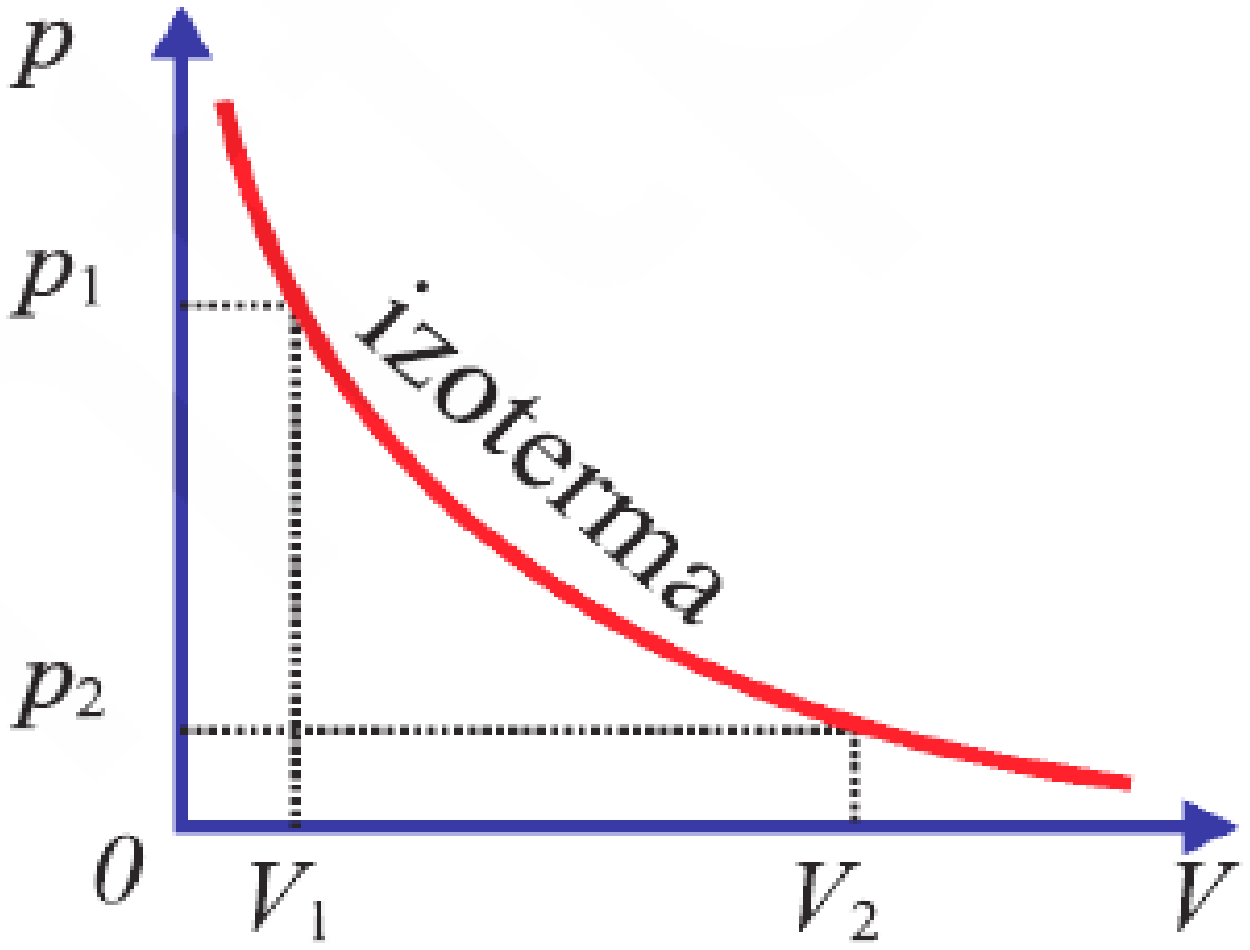
Izotermik jarayon

$$T = const.$$

$$p \cdot V = const.$$

$$p_1 \cdot V_1 = p_2 \cdot V_2$$

$$\frac{p_1}{p_2} = \frac{V_2}{V_1}$$

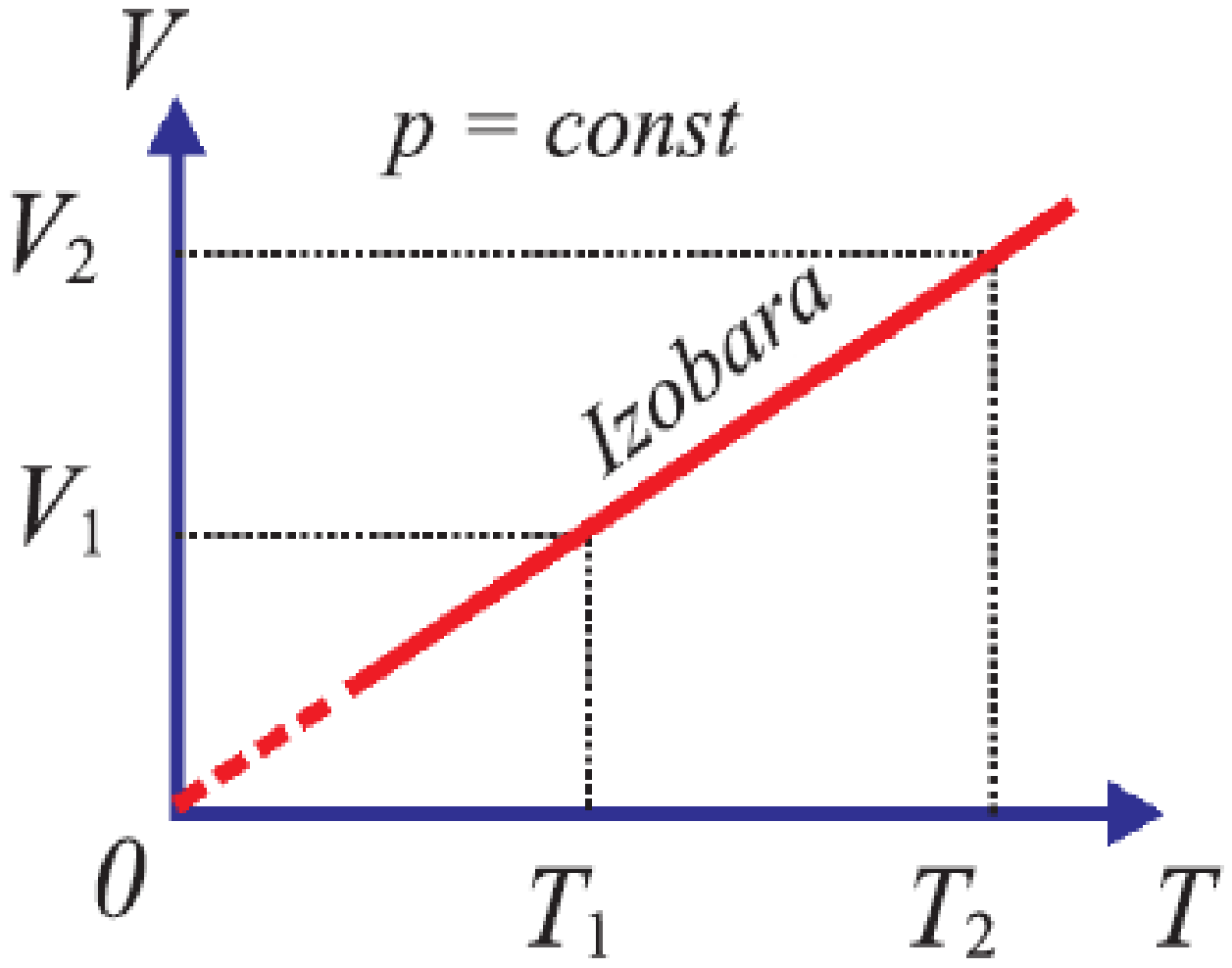


Izobarikik jarayon

$$\mathrm{p} = const.$$

$$\frac{V}{T} = const.$$

$$\frac{V_1}{V_2} = \frac{T_1}{T_2}$$

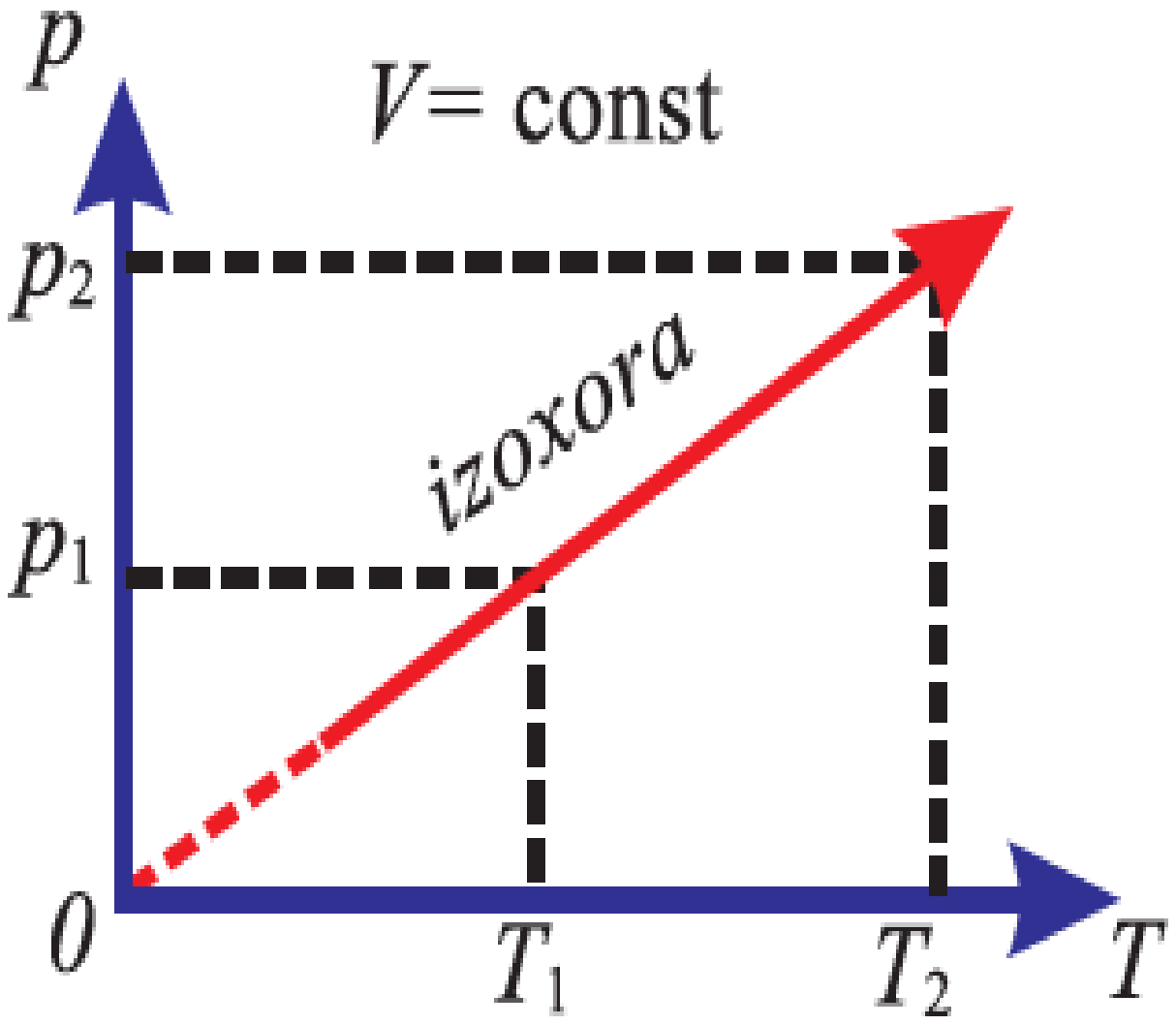


Izoxorik jarayon

$$\mathrm{V} = const.$$

$$\frac{P}{T} = const.$$

$$\frac{p_1}{p_2} = \frac{T_1}{T_2}$$



1-masala

Ideal gaz 6 l hajmdan 4 l hajmgacha izotermik siqildi, bunda gazning bosimi 0,6 normal atmosfera bosimiga ortdi. Gazning dastlabki bosimi qanday bo‘lgan? Normal atmosfera bosimini 100 kPa deb oling.

Berilgan:

$V_1 = 6\ l$

$V_2 = 4\ l$

$\Delta p = 0{,}6 \cdot p_A$

$p_A = 100\ kPa$

$T = const.$

Topish kerak:

$p_1 - ?$

Formula:

$T = const.$ da $\quad p_1 \cdot V_1 = p_2 \cdot V_2$

$$p_2 = p_1 + \Delta p$$

$$p_1 \cdot V_1 = (p_1 + \Delta p) \cdot V_2$$

$$p_1 \cdot V_1 = p_1 \cdot V_2 + \Delta p \cdot V_2$$

$$p_1 \cdot V_1 - p_1 \cdot V_2 = \Delta p \cdot V_2$$

$$p_1 \cdot (V_1 - V_2) = \Delta p \cdot V_2$$

$$p_1 = \frac{\Delta p \cdot V_2}{(V_1 - V_2)}$$

$$p_1 = \frac{\Delta p \cdot V_2}{(V_1 - V_2)} = \frac{0{,}6 \cdot p_A \cdot V_2}{(V_1 - V_2)}$$

Yechish:

$$p_1 = \frac{0{,}6 \cdot 100\ kPa \cdot 4\ l}{6\ l - 4\ l} = 120\ kPa$$

Javob: $p_1 = 120\ kPa$

2- masala

11-mashq 2-masala

27℃ temperaturada yopiq idishdagi gazning bosimi 900 kPa ga teng bo‘lgan. Gaz qizdirilib temperaturasi 227℃ ga yetkazilganda, idish ichidagi gazning bosimi qanday qiymatga erishadi?

Berilgan:

$t_1 = 27℃$

$V = const.$

$p_1 = 900\ kPa$

$t_2 = 227℃$

Formula:

$V = const$ da $\frac{p_1}{p_2} = \frac{T_1}{T_2}$

$$p_2 = \frac{p_1 \cdot T_2}{T_1}$$

$$T = t + 273$$

Topish kerak: $p_2 - ?$

Yechish: $T_1 = 27℃ + 273 = 300\ K$

$T_2 = 227℃ + 273 = 500\ K$

$$p_2 = \frac{900\ kPa \cdot 500\ K}{300\ K} = 1500\ kPa = 1{,}5\ MPa$$

Javob: $p_2 = 1,5\ MPa$

3-masala

11-mashq 3-masala

Dastlabki temperaturasi 27℃ bo‘lgan ideal gaz izobarik kengayib, uning hajmi 24 % ga ortdi. Uning keyingi temperaturasi qanday bo‘lgan?

Berilgan:

$T_1 = 27℃ + 273 = 300\ K$

$\Delta V = \frac{24\ \%}{100\ \%} \cdot V_1$

$p = const.$

Topish kerak: $T_2 - ?$

Formula:

$p = const.$

$\frac{V_1}{V_2} = \frac{T_1}{T_2}$

$V_2 = V_1 + \Delta V = V_1 + \frac{24\ \%}{100\ \%} \cdot V_1$

$V_2 = V_1 + 0{,}24 \cdot V_1 = 1{,}24 \cdot V_1$

$$\frac{V_1}{1{,}24 \cdot V_1} = \frac{T_1}{T_2} \Rightarrow \frac{1}{1{,}24} = \frac{T_1}{T_2} \Rightarrow T_2 = 1{,}24 \cdot T_1$$

Yechish: $T_2 = 1{,}24 \cdot 300\ K = 372\ K$

Javob: $\boldsymbol{T_2 = 372\ K}$

5-masala

11-mashq 5-masala

Havo puffakchasi suv havzasining tubidan yuziga chiqquncha 3,5 marta kattalshadi. Suv havzasining chuqurligi qanday? Suvning ustki va pastki qismida temperaturalar bir xil deb hisoblang.

Berilgan:

$\frac{V_2}{V_1} = 3{,}5$

$P_A = 100000\ Pa$

$g = 10\ m/s^2$

$\rho = 1000\ kg/m^3$

$T = const.$

Topish kerak:

$h - ?$

Formula va yechish:

$$T = const$$

$$\frac{p_1}{p_2} = \frac{V_2}{V_1} \Rightarrow \frac{p_1}{p_2} = 3{,}5$$

$$P_1 = P_A + \rho \cdot g \cdot h \qquad P_2 = P_A$$

$$\frac{P_A + \rho \cdot g \cdot h}{P_A} = 3{,}5$$

$$P_A + \rho \cdot g \cdot h = 3{,}5 \cdot P_A$$

$$\rho \cdot g \cdot h = 2{,}5 \cdot P_A \qquad h = \frac{2{,}5 \cdot P_A}{\rho \cdot g}$$

$$h = \frac{2{,}5 \cdot P_A}{\rho \cdot g}$$

$$h = \frac{2{,}5 \cdot 100000\ Pa}{1000\frac{kg}{m^3} \cdot 10\frac{m}{s^2}} = 25\ m$$

$$[h] = \frac{Pa}{\frac{kg}{m^3} \cdot \frac{m}{s^2}} = \frac{N/m^2}{N/m^3} = \text{m}$$

Javob: $h = 25\ m$

Mustaqil bajarish uchun topshiriqlar

11-mashqning 2,4,6-masalalarini yechish.